Oppg 1 – Fourier transform

$$\hat{f}(\omega) = \mathcal{F}(f) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

a) $f(x) = e^{-2|x|}$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-2|x|} \cdot e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^{0} e^{2x - i\omega x} dx + \int_{0}^{\infty} e^{-2x - i\omega x} dx \right)$$

$$= \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^{0} e^{x(2 - i\omega)} dx + \int_{0}^{\infty} e^{-x(2 + i\omega)} dx \right)$$

$$= \frac{1}{\sqrt{2\pi}} \left(\left[\frac{e^{x(2-i\omega)}}{2 - i\omega} \right]_{-\infty \Rightarrow 0}^{0} - \left[\frac{e^{-x(2+i\omega)}}{2 + i\omega} \right]_{0}^{\infty \Rightarrow 0} \right)$$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{2 - i\omega} + \frac{1}{2 + i\omega} \right)$$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{2 + i\omega}{4 + \omega^2} + \frac{2 - i\omega}{4 + \omega^2} \right)$$

$$= \underline{\sqrt{\frac{2}{\pi}} \frac{2}{4 + \omega^2}}$$

b) $f(x) = e^{x}, \quad x \in [-a, a]$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-a}^{a} e^{x} \cdot e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-a}^{a} e^{x(1 - i\omega)} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{x(1-i\omega)}}{1 - i\omega} \right]_{-a}^{a}$$

$$= \underline{\frac{1}{\sqrt{2\pi}} \left(\frac{e^{a(1-i\omega)} - e^{-a(1-i\omega)}}{1 - i\omega} \right)}$$

c) $f(x) = x^2 e^{-x^2}$

$$F\left(\frac{df}{dx}\right) = i\omega F(f) \Rightarrow F\left(\frac{d^2 f}{dx^2}\right) = -\omega^2 F(f)$$

$$g(x) = e^{-x^2} \Rightarrow g'(x) = -2x e^{-x^2} \Rightarrow g''(x) = 4x^2 e^{-x^2} - 2e^{-x^2}$$

$$g''(x) = 4f(x) - 2e^{-x^2} \Rightarrow f(x) = \frac{1}{4} g''(x) + \frac{1}{2} e^{-x^2} = \frac{1}{4} \frac{d^2 e^{-x^2}}{dx^2} + \frac{1}{2} e^{-x^2}$$

$$F(f) = \frac{1}{4} F\left(\frac{d^2 e^{-x^2}}{dx^2}\right) + \frac{1}{2} F(e^{-x^2})$$

$$= -\frac{1}{4} \omega^2 F(e^{-x^2}) + \frac{1}{2} F(e^{-x^2}), \quad F(e^{-x^2}) = \frac{1}{\sqrt{2}} e^{-\frac{\omega^2}{4}}$$

$$= e^{-\frac{\omega^2}{4}} \left(\frac{1}{2\sqrt{2}} - \frac{\omega^2}{4\sqrt{2}} \right)$$

Oppg 2 – Convolution

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x - \tau) g(\tau) d\tau \Rightarrow \mathcal{F}(f * g)(x) = \sqrt{2\pi}\, \mathcal{F}(f) \mathcal{F}(g)$$

$$f(x) = e^{-x^2} \Rightarrow \hat{f}(\omega) = \frac{1}{\sqrt{2}} e^{-\frac{\omega^2}{4}}$$

$$g(x) = e^{-2x^2} \Rightarrow \hat{g}(\omega) = \frac{1}{2} e^{-\frac{\omega^2}{8}}$$

$$(f*g)(x) = \sqrt{2\pi}\, \mathcal{F}^{-1}(\hat{f} \cdot \hat{g})$$

$$= \frac{\sqrt{\pi}}{2} \mathcal{F}^{-1}\left(e^{-\frac{\omega^2}{4}(1+\frac{1}{2})}\right)$$

$$= \frac{\sqrt{\pi}}{2} \mathcal{F}^{-1}\left(e^{-\frac{3\omega^2}{8}}\right)$$

$$\hat{f}(\omega) = e^{-\frac{\omega^2}{4a}} = \sqrt{2a}\, e^{-ax^2} = f(x) = \mathcal{F}^{-1}(\hat{f})$$

$$(f*g)(x) = \frac{\sqrt{\pi}}{2} \mathcal{F}^{-1}\left(e^{-\frac{3\omega^2}{8}}\right), \quad a = \frac{2}{3}$$

$$= \frac{\sqrt{\pi}}{2} \sqrt{2 \cdot \frac{2}{3}}\; e^{-\frac{2}{3}x^2}$$

$$= \underline{\sqrt{\frac{\pi}{3}}\, e^{-\frac{2}{3}x^2}}$$

Oppg 3 - Convolution

$$f(x) := \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{else} \end{cases} \qquad g(x) := \begin{cases} x, & 0 < x < 1 \\ 2-x, & 1 < x < 2 \\ 0, & \text{else} \end{cases}$$

a) $(f*f)(x) = \int_{-\infty}^{\infty} f(y) f(x-y)\, dy$

$$f(x-y) = \begin{cases} 1, & 0 < x-y < 1 \\ 0, & \text{else} \end{cases}$$

$$(f*f)(x) = \int_{x-1}^{x} f(y)\, dy$$

if $x \leq 0$: $(f*f)(x) = 0$

if $0 < x \leq 1$: $(f*f)(x) = x$

if $1 < x \leq 2$: $(f*f)(x) = 2-x$

if $x > 2$: $(f*f)(x) = 0$

b) $\hat{f}(\omega) = \mathcal{F}(f) = \frac{1}{\sqrt{2\pi}} \int_0^1 1\, e^{-i\omega x}\, dx$

$$= \frac{1}{\sqrt{2\pi}} \left[-\frac{e^{-i\omega x}}{i\omega} \right]_0^1$$

$$= \frac{1}{\sqrt{2\pi}} \left(-\frac{e^{-i\omega}}{i\omega} - -\frac{e^{-i\omega 0}}{i\omega} \right)$$

$$= \underline{\frac{1 - e^{-i\omega}}{\sqrt{2\pi}\, i\omega}}$$

c) $\hat{g}(\omega) = \mathcal{F}(g) = \mathcal{F}(f*f) = \sqrt{2\pi}\, \mathcal{F}(f)\mathcal{F}(f) = \sqrt{2\pi}\left(\frac{1-e^{-i\omega}}{\sqrt{2\pi}\, i\omega}\right)^2 = \underline{-\frac{(1-e^{-i\omega})^2}{\sqrt{2\pi}\, \omega^2}}$

Oppg 4 - DFT

a) if $j \bmod N = 0 \Rightarrow j = mN,\ m \in \mathbb{Z}$

$$\omega_N^j = e^{\frac{-2\pi j i}{N}} = e^{-2\pi m i} = 1^m = 1 \implies \sum_{k=0}^{N-1} \omega_N^{jk} = \sum_{k=0}^{N-1} 1^k = N$$

if $j \bmod N \neq 0$

$$\sum_{k=0}^{N-1} \omega_N^{jk} = \frac{1 - \overset{=1}{\omega_N^{Nj}}}{1 - \omega_N^j} = 0$$

b) if $n = m$:

$$\sum_{k=0}^{N-1} e^{\frac{2\pi i n k}{N}} e^{-\frac{2\pi i m k}{N}} = \sum_{k=0}^{N-1} e^{\frac{2\pi i k}{N}\overset{0}{\overset{\shortparallel}{(n-m)}}} = \sum_{k=0}^{N-1} 1 = N$$

if $n \neq m$: see pt.2 in a)

c) $\mathcal{F}_N = \left(e^{\frac{-2\pi i j n}{N}}\right)_{j,n=0,\dots,N-1}$

$$\mathcal{F}_2 = \begin{bmatrix} e^{\frac{-2\pi i 0\cdot 0}{2}} & e^{\frac{-2\pi i 0\cdot 1}{2}} \\ e^{\frac{-2\pi i 1\cdot 0}{2}} & e^{\frac{-2\pi i 1\cdot 1}{2}} \end{bmatrix} = \begin{bmatrix} e^0 & e^0 \\ e^0 & e^{-\pi i} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\mathcal{F}_3 = \begin{bmatrix} e^{\frac{-2\pi i 0\cdot 0}{3}} & e^{\frac{-2\pi i 0\cdot 1}{3}} & e^{\frac{-2\pi i 0\cdot 2}{3}} \\ e^{\frac{-2\pi i 1\cdot 0}{3}} & e^{\frac{-2\pi i 1\cdot 1}{3}} & e^{\frac{-2\pi i 1\cdot 2}{3}} \\ e^{\frac{-2\pi i 2\cdot 0}{3}} & e^{\frac{-2\pi i 2\cdot 1}{3}} & e^{\frac{-2\pi i 2\cdot 2}{3}} \end{bmatrix} = \begin{bmatrix} e^0 & e^0 & e^0 \\ e^0 & e^{\frac{-2\pi i}{3}} & e^{\frac{-4\pi i}{3}} \\ e^0 & e^{\frac{-4\pi i}{3}} & e^{\frac{-8\pi i}{3}} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \frac{1}{2} - \frac{\sqrt{3}}{2} i & \frac{1}{2} + \frac{\sqrt{3}}{2} i \\ 1 & \frac{1}{2} + \frac{\sqrt{3}}{2} i & \frac{1}{2} - \frac{\sqrt{3}}{2} i \end{bmatrix}$$

$$\mathcal{F}_4 = \begin{bmatrix} e^{\frac{-2\pi i 0\cdot 0}{4}} & e^{\frac{-2\pi i 0\cdot 1}{4}} & e^{\frac{-2\pi i 0\cdot 2}{4}} & e^{\frac{-2\pi i 0\cdot 3}{4}} \\ e^{\frac{-2\pi i 1\cdot 0}{4}} & e^{\frac{-2\pi i 1\cdot 1}{4}} & e^{\frac{-2\pi i 1\cdot 2}{4}} & e^{\frac{-2\pi i 1\cdot 3}{4}} \\ e^{\frac{-2\pi i 2\cdot 0}{4}} & e^{\frac{-2\pi i 2\cdot 1}{4}} & e^{\frac{-2\pi i 2\cdot 2}{4}} & e^{\frac{-2\pi i 2\cdot 3}{4}} \\ e^{\frac{-2\pi i 3\cdot 0}{4}} & e^{\frac{-2\pi i 3\cdot 1}{4}} & e^{\frac{-2\pi i 3\cdot 2}{4}} & e^{\frac{-2\pi i 3\cdot 3}{4}} \end{bmatrix} = \begin{bmatrix} e^0 & e^0 & e^0 & e^0 \\ e^0 & e^{\frac{-\pi i}{2}} & e^{-\pi i} & e^{-\frac{3\pi i}{2}} \\ e^0 & e^{-\pi i} & e^{-2\pi i} & e^{-3\pi i} \\ e^0 & e^{-\frac{3\pi i}{2}} & e^{-3\pi i} & e^{-\frac{9\pi i}{4}} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}$$

d) $f = [1, 2, 3, 3]^T$

$$\hat{f} = \mathcal{F}_4 f \implies \hat{f} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 3 \\ 1 & -2i & -3 & 3i \\ 1 & -2 & 3 & -3 \\ 1 & 2i & -3 & -3i \end{bmatrix} = \begin{bmatrix} 9 \\ i-2 \\ -1 \\ -i-2 \end{bmatrix}$$